

## 1.2 Rank, column space and null space

### 1.2.1 Subspaces and dimension

A nonempty set  $\mathcal{S} \subseteq \mathbb{R}^n$  is a **subspace** if  $\mathbf{u}, \mathbf{v} \in \mathcal{S}$  and  $a, b \in \mathbb{R}$  imply  $a\mathbf{u} + b\mathbf{v} \in \mathcal{S}$ . Vectors  $\mathbf{v}_1, \dots, \mathbf{v}_k$  are **linearly independent** if  $\sum_j c_j \mathbf{v}_j = \mathbf{0}$  only for  $c_1 = \dots = c_k = 0$ , that is, if the matrix  $[\mathbf{v}_1, \dots, \mathbf{v}_k]$  sends only the zero vector to zero. Their **span** is the set of all their linear combinations. A **basis** of  $\mathcal{S}$  is an independent set that spans it.

We use the following facts of elementary linear algebra without proof (Halmos 1958, chapter 1). Every subspace of  $\mathbb{R}^n$  has a basis. All bases of a subspace have the same number of elements, its **dimension**  $\dim \mathcal{S}$ . In an  $r$ -dimensional subspace, any  $r$  independent vectors form a basis, and any independent set can be extended to a basis. If  $\mathcal{S} \subseteq \mathcal{T}$  and  $\dim \mathcal{S} = \dim \mathcal{T}$ , then  $\mathcal{S} = \mathcal{T}$ . Finally, if  $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$ , the union of a basis of  $\mathcal{S}$  and a basis of  $\mathcal{T}$  is independent, so the sum  $\mathcal{S} + \mathcal{T} = \{\mathbf{u} + \mathbf{v} : \mathbf{u} \in \mathcal{S}, \mathbf{v} \in \mathcal{T}\}$  has dimension  $\dim \mathcal{S} + \dim \mathcal{T}$ . Such a sum is called **direct** and written  $\mathcal{S} \oplus \mathcal{T}$ .

### 1.2.2 The four subspaces of a matrix

An  $m \times n$  matrix  $\mathbf{A}$  carries two subspaces of  $\mathbb{R}^m$  and two of  $\mathbb{R}^n$ :

$$\begin{aligned} \mathcal{C}(\mathbf{A}) &= \{\mathbf{A}\mathbf{b} : \mathbf{b} \in \mathbb{R}^n\} \subseteq \mathbb{R}^m, & \mathcal{N}(\mathbf{A}) &= \{\mathbf{b} \in \mathbb{R}^n : \mathbf{A}\mathbf{b} = \mathbf{0}\} \subseteq \mathbb{R}^n, \\ \mathcal{C}(\mathbf{A}^\top) &\subseteq \mathbb{R}^n, & \mathcal{N}(\mathbf{A}^\top) &\subseteq \mathbb{R}^m. \end{aligned}$$

By Eq. 1.1.2,  $\mathcal{C}(\mathbf{A})$  is the span of the columns, the **column space**, and  $\mathcal{C}(\mathbf{A}^\top)$  is the span of the rows, the **row space**.  $\mathcal{N}(\mathbf{A})$  is the **null space**. The **rank** of  $\mathbf{A}$  is  $\text{rank}(\mathbf{A}) = \dim \mathcal{C}(\mathbf{A})$ , the largest number of independent columns. The number of independent rows,  $\dim \mathcal{C}(\mathbf{A}^\top)$ , looks like a different quantity. It is not.

#### Theorem 1.2.1 — Row rank equals column rank

For every matrix  $\mathbf{A}$ ,  $\dim \mathcal{C}(\mathbf{A}^\top) = \dim \mathcal{C}(\mathbf{A})$ . That is,  $\text{rank}(\mathbf{A}^\top) = \text{rank}(\mathbf{A})$ . Moreover, if  $\text{rank}(\mathbf{A}) = r \geq 1$  there are an  $m \times r$  matrix  $\mathbf{B}$  and an  $r \times n$  matrix  $\mathbf{F}$ , both of rank  $r$ , with  $\mathbf{A} = \mathbf{B}\mathbf{F}$ .

*Proof.* Let  $r = \text{rank}(\mathbf{A})$  and let the columns of the  $m \times r$  matrix  $\mathbf{B}$  be a basis of  $\mathcal{C}(\mathbf{A})$ . Each column  $\mathbf{a}_j$  is a combination  $\mathbf{B}\mathbf{f}_j$  of these, so  $\mathbf{A} = \mathbf{B}\mathbf{F}$  with  $\mathbf{F} = [\mathbf{f}_1, \dots, \mathbf{f}_n]$ . By the transposed reading of Eq. 1.1.2, every row of  $\mathbf{A} = \mathbf{B}\mathbf{F}$  is a combination of the  $r$  rows of  $\mathbf{F}$ , so  $\dim \mathcal{C}(\mathbf{A}^\top) \leq r = \dim \mathcal{C}(\mathbf{A})$ . Applying this to  $\mathbf{A}^\top$  gives the reverse inequality. Finally  $\text{rank}(\mathbf{F}) \leq r$  because  $\mathbf{F}$  has  $r$  rows, and  $r = \dim \mathcal{C}(\mathbf{A}^\top) \leq \dim \mathcal{C}(\mathbf{F}^\top) = \text{rank}(\mathbf{F})$ , so  $\text{rank}(\mathbf{F}) = r$ . ■

The factorization  $\mathbf{A} = \mathbf{B}\mathbf{F}$  is a **rank factorization**:  $\mathbf{B}$  has **full column rank** (its columns are independent) and  $\mathbf{F}$  has **full row rank**. It is a convenient tool for proofs, as Proposition 1.9.1 shows.

### Theorem 1.2.2 — Rank–nullity

For an  $m \times n$  matrix  $\mathbf{A}$ ,  $\text{rank}(\mathbf{A}) + \dim \mathcal{N}(\mathbf{A}) = n$ .

*Proof.* Let  $\mathbf{v}_1, \dots, \mathbf{v}_k$  be a basis of  $\mathcal{N}(\mathbf{A})$  and extend it to a basis  $\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w}_1, \dots, \mathbf{w}_{n-k}$  of  $\mathbb{R}^n$ . Every  $\mathbf{A}\mathbf{b}$  is a combination of  $\mathbf{A}\mathbf{w}_1, \dots, \mathbf{A}\mathbf{w}_{n-k}$ , since  $\mathbf{A}\mathbf{v}_i = \mathbf{0}$ . These vectors are independent: if  $\sum_j c_j \mathbf{A}\mathbf{w}_j = \mathbf{0}$ , then  $\sum_j c_j \mathbf{w}_j \in \mathcal{N}(\mathbf{A})$  is a combination of the  $\mathbf{v}_i$ , and independence of the full basis forces every  $c_j = 0$ . So  $\text{rank}(\mathbf{A}) = n - k$ . ■

In a linear model with  $n \times p$  model matrix  $\mathbf{X}$ ,  $\dim \mathcal{N}(\mathbf{X}) = p - \text{rank}(\mathbf{X})$  counts the directions in which the coefficient vector can move without changing the mean  $\mathbf{X}\boldsymbol{\beta}$ .  $\mathbf{X}$  has full column rank iff  $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$ .

## 1.2.3 Rank of products and of $\mathbf{A}^\top \mathbf{A}$

### Proposition 1.2.3

Let  $\mathbf{A}$  be  $m \times n$  and  $\mathbf{B}$  be  $n \times q$ .

- (a)  $\mathcal{C}(\mathbf{AB}) \subseteq \mathcal{C}(\mathbf{A})$  and  $\mathcal{N}(\mathbf{B}) \subseteq \mathcal{N}(\mathbf{AB})$ , so  $\text{rank}(\mathbf{AB}) \leq \min\{\text{rank}(\mathbf{A}), \text{rank}(\mathbf{B})\}$ .
- (b) If  $\mathbf{B}$  has full row rank, then  $\mathcal{C}(\mathbf{AB}) = \mathcal{C}(\mathbf{A})$ . If  $\mathbf{A}$  has full column rank, then  $\mathcal{N}(\mathbf{AB}) = \mathcal{N}(\mathbf{B})$ . So  $\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{A})$  in the first case and  $\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{B})$  in the second. In particular, multiplying by a nonsingular matrix does not change rank.
- (c)  $\mathcal{N}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A})$  and  $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)$ , so  $\text{rank}(\mathbf{A}^\top \mathbf{A}) = \text{rank}(\mathbf{AA}^\top) = \text{rank}(\mathbf{A})$ .
- (d)  $\text{rank}(\mathbf{A} + \mathbf{C}) \leq \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{C})$  for  $\mathbf{C}$  of the same size as  $\mathbf{A}$ .

*Proof.*

- (a) is Eq. 1.1.2, together with  $\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{B}^\top \mathbf{A}^\top) \leq \text{rank}(\mathbf{B}^\top)$  by the first inclusion and Theorem 1.2.1. (b) If  $\mathbf{B}$  has full row rank,  $\mathcal{C}(\mathbf{B}) = \mathbb{R}^n$  because it is an  $n$ -dimensional subspace of  $\mathbb{R}^n$ , so every  $\mathbf{A}\mathbf{u}$  equals  $\mathbf{AB}\mathbf{v}$  for some  $\mathbf{v}$ . If  $\mathbf{A}$  has full column rank,  $\mathbf{AB}\mathbf{v} = \mathbf{0}$  forces  $\mathbf{B}\mathbf{v} = \mathbf{0}$ . The rank statements follow from rank–nullity. (c) If  $\mathbf{A}^\top \mathbf{A}\mathbf{v} = \mathbf{0}$  then  $\|\mathbf{A}\mathbf{v}\|^2 = \mathbf{v}^\top \mathbf{A}^\top \mathbf{A}\mathbf{v} = 0$ , so  $\mathbf{A}\mathbf{v} = \mathbf{0}$ . Thus the null spaces agree, and by Theorem 1.2.2  $\text{rank}(\mathbf{A}^\top \mathbf{A}) = \text{rank}(\mathbf{A})$ . Since  $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) \subseteq \mathcal{C}(\mathbf{A}^\top)$  and both have dimension  $\text{rank}(\mathbf{A})$ , they are equal. Applying this to  $\mathbf{A}^\top$  gives  $\text{rank}(\mathbf{AA}^\top) = \text{rank}(\mathbf{A})$ . (d)  $\mathcal{C}(\mathbf{A} + \mathbf{C}) \subseteq \mathcal{C}(\mathbf{A}) + \mathcal{C}(\mathbf{C})$ , and the dimension of a sum is at most the sum of the dimensions. ■

Part (c) is the reason the normal equations  $\mathbf{X}^\top \mathbf{X}\mathbf{b} = \mathbf{X}^\top \mathbf{y}$  always have a solution: the right-hand side lies in  $\mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{X})$  (Section 1.4).

## 1.2.4 Nonsingular submatrices

### Proposition 1.2.4

Let  $\mathbf{A}$  have rank  $r \geq 1$ . Choose any  $r$  linearly independent rows, indexed by  $I$ , and any  $r$  linearly independent columns, indexed by  $J$ . Then the  $r \times r$  submatrix  $\mathbf{A}_{IJ}$  at their intersection is nonsingular, and

$$\mathbf{A} = \mathbf{A}_{.J} \mathbf{A}_{IJ}^{-1} \mathbf{A}_{I.}, \quad (1.2.1)$$

where  $\mathbf{A}_{.J}$  holds the columns in  $J$  and  $\mathbf{A}_{I.}$  the rows in  $I$ .

*Proof.* The columns in  $J$  are a basis of  $\mathcal{C}(\mathbf{A})$ , so  $\mathbf{A} = \mathbf{A}_{.J} \mathbf{F}$  for some  $r \times n$  matrix  $\mathbf{F}$ . Keeping only the rows in  $I$  gives  $\mathbf{A}_{I.} = \mathbf{A}_{IJ} \mathbf{F}$ . The left side has rank  $r$ , so by Proposition 1.2.3(a) the  $r \times r$  matrix  $\mathbf{A}_{IJ}$  has rank  $r$  and is nonsingular. Then  $\mathbf{F} = \mathbf{A}_{IJ}^{-1} \mathbf{A}_{I.}$ , which gives Eq. 1.2.1. ■

Conversely, a nonsingular  $k \times k$  submatrix makes the  $k$  columns of  $\mathbf{A}$  that contain it linearly independent, so  $\text{rank}(\mathbf{A}) \geq k$ . In particular, a matrix has rank  $r$  iff it has a nonsingular  $r \times r$  submatrix and no nonsingular larger one. This proposition gives an explicit generalized inverse in Theorem 1.3.2.

## 1.2.5 Exercises

### 1.2.5.1 A. Check your understanding

#### Exercise A1.

Show that  $\sum_{i < j} (y_i - y_j)^2 = n \sum_i (y_i - \bar{y})^2$ , and write both sides as  $\mathbf{y}^\top \mathbf{A} \mathbf{y}$  with  $\mathbf{A}$  symmetric. Find the symmetric matrix of the quadratic form  $2x_1^2 - x_2^2 + 6x_1x_2 - 4x_2x_3$ .

#### Exercise A2.

Show that  $\mathbf{X}^\top \mathbf{X} \mathbf{B} = \mathbf{X}^\top \mathbf{X} \mathbf{C}$  implies  $\mathbf{X} \mathbf{B} = \mathbf{X} \mathbf{C}$ . Deduce that  $\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{0}$  iff  $\mathbf{X} \mathbf{b} = \mathbf{0}$ .

*Solution.* Put  $\mathbf{D} = \mathbf{B} - \mathbf{C}$ , so that  $\mathbf{X}^\top \mathbf{X} \mathbf{D} = \mathbf{0}$ . Then  $(\mathbf{X} \mathbf{D})^\top (\mathbf{X} \mathbf{D}) = \mathbf{D}^\top \mathbf{X}^\top \mathbf{X} \mathbf{D} = \mathbf{0}$ , and by Eq. 1.1.3 a matrix  $\mathbf{Z}$  with  $\mathbf{Z}^\top \mathbf{Z} = \mathbf{0}$  is zero. So  $\mathbf{X} \mathbf{D} = \mathbf{0}$ . The second statement is the case  $\mathbf{C} = \mathbf{0}$ , with the converse obvious.

#### Exercise A3.

For the additive two-way layout with two rows, three columns and one observation per cell, write the  $6 \times 6$  model matrix  $\mathbf{X} = [\mathbf{1}, \mathbf{R}, \mathbf{C}]$ , where  $\mathbf{R}$  and  $\mathbf{C}$  hold the row and column indicators. Find  $\text{rank}(\mathbf{X})$  and a basis of  $\mathcal{N}(\mathbf{X})$ . Compare  $[\mathbf{R}, \mathbf{C}]^\top$  with the

matrix  $\mathbf{A}$  of Example (Filling a table from its margins).

### 1.2.5.2 B. Practice

#### Exercise B1.

Let  $\mathbf{A}$  be  $m \times n$  and  $\mathbf{B}$  be  $n \times q$ . Prove  $\text{rank}(\mathbf{AB}) \geq \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B}) - n$ . *Hint:* apply rank-nullity to  $\mathbf{AB}_0$ , where the columns of  $\mathbf{B}_0$  are a basis of  $\mathcal{C}(\mathbf{B})$ .

*Solution.* Let the  $n \times s$  matrix  $\mathbf{B}_0$ ,  $s = \text{rank}(\mathbf{B})$ , have columns forming a basis of  $\mathcal{C}(\mathbf{B})$ . Then  $\mathcal{C}(\mathbf{AB}) = \mathcal{C}(\mathbf{AB}_0)$ , so by rank-nullity  $\text{rank}(\mathbf{AB}) = s - \dim \mathcal{N}(\mathbf{AB}_0)$ . The map  $\mathbf{c} \mapsto \mathbf{B}_0 \mathbf{c}$  is one-to-one and sends  $\mathcal{N}(\mathbf{AB}_0)$  into  $\mathcal{N}(\mathbf{A})$ , so  $\dim \mathcal{N}(\mathbf{AB}_0) \leq \dim \mathcal{N}(\mathbf{A}) = n - \text{rank}(\mathbf{A})$ . Hence  $\text{rank}(\mathbf{AB}) \geq \text{rank}(\mathbf{B}) - n + \text{rank}(\mathbf{A})$ .

#### Exercise B2.

Show that if  $\mathbf{A} = \mathbf{B}_1 \mathbf{F}_1 = \mathbf{B}_2 \mathbf{F}_2$  are two rank factorizations, then  $\mathbf{B}_2 = \mathbf{B}_1 \mathbf{T}$  and  $\mathbf{F}_2 = \mathbf{T}^{-1} \mathbf{F}_1$  for a nonsingular  $\mathbf{T}$ .